

Testing Plan for Interactive Prototype 2

1 Project Background

This project extends the XR redesign of a creative drawing app (inspired by Procreate) into a functional VR prototype on the Meta Quest headset. Unlike Prototype 1 (mouse/keyboard simulation), this version features **fully interactive 3D models inside VR**:

- A **pen tool represented by a 3D pen model**, held and used naturally in VR.
- Erasers with multiple effects for correction tasks.
- **Physical-like buttons** with colliders and audio feedback that trigger actions (e.g., clear/reset).
- A **wrist menu** opened with the controller menu button (functions not yet active).
- **Remote grab functionality**: tools (pen, eraser, weapon models) can be grabbed from a distance and, when released, float in place without gravity.

The aim is to evaluate how immersive XR interactions—drawing, erasing, button pressing, wrist menu, and remote grabbing—affect usability, realism, and convenience.

2 Testing Objectives

- Test how easy and natural it feels to **draw with the 3D pen model** in VR (smoothness, accuracy, comfort).
- Evaluate **eraser functions**: whether different effects are intuitive, responsive, and effective.

- Assess usability of **physical buttons with collision and audio feedback**: can participants press them easily, and do they feel realistic?
 - Check whether the **wrist menu** can be reliably opened using the controller's menu button, and whether its position feels appropriate.
 - Test **remote grab & floating tools**: whether grabbing/releasing tools without gravity is convenient, or if users prefer tools to drop naturally.
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3 Testing Methodology

- **Think-Aloud Protocol**: Participants describe their thoughts while drawing, erasing, pressing buttons, opening menus, and using remote grab.
 - **Task-Based Testing**: Assign 5 tasks:
 1. Use the 3D pen model to draw a simple shape (e.g., circle).
 2. Erase part of the drawing using two different eraser effects.
 3. Press the physical “clear/reset” button to reset the canvas and hear audio feedback.
 4. Open the wrist menu using the controller's menu button and comment on its placement.
 5. Use remote grab to pick up the pen or eraser from a distance, then release it to float in place. Comment on whether this feels useful.
 - **Oral Inquiry**: Ask participants how they imagine tool behavior (float vs drop) and future wrist menu functions.
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4 Prototype Description / Requirements

The prototype is implemented in Unity and runs on Meta Quest. Key components include:

- **3D pen model:** held in VR, allows freehand drawing on the canvas.
 - **Erasers:** with multiple modes (e.g., soft erase, hard erase).
 - **Physical buttons:** collider-based, pressable, with audio and functional feedback.
 - **Wrist menu:** opened via controller menu button, appears anchored to the wrist (non-functional placeholder).
 - **Remote grab:** allows users to pull tools from a distance; when released, tools float in position instead of falling.
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5 Data Collection & Success Criteria

a) Data Collection

- *Quantitative:* Success rate of tasks (drawing, erasing, button pressing, menu opening, remote grab), number of errors.
- *Qualitative:* User comments on drawing smoothness, eraser satisfaction, button realism, wrist menu comfort, and tool floating.
- *Ratings:* 1–5 for satisfaction with each interaction type (pen, eraser, button, menu, remote grab).

b) Success Criteria

- $\geq 80\%$ of participants complete all tasks without external help.
- Average satisfaction $\geq 4/5$ for pen and button interactions.
- At least 70% of participants find eraser and remote grab convenient and effective.
- Participants provide clear feedback on wrist menu and tool-floating usefulness.

6 Testing Setup

- Unity XR prototype running on **Meta Quest headset**.
 - Quiet and safe VR space.
 - Researcher present to guide participants and record data.
 - Optional screen capture for analysis.
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7 Testing Process (6–8 min per participant)

1. **Intro & consent (30 sec):** Explain purpose, assure participants they are testing the prototype—not themselves.
2. **Warm-up & demo (30–40 sec):** Show how to hold pen, erase, press button, open wrist menu, and use remote grab.
3. **Task execution (4–5 min):** Participants perform the five assigned tasks.
4. **Debrief & interview (1–2 min):** Ask about comfort, usability, frustrations, and preferences (e.g., floating vs dropping tools).